



Applied Containerization for Machine Learning in HPC

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- Website: <http://www.colorado.edu/rc>
- Documentation: <https://curc.readthedocs.io>
- Helpdesk: rc-help@colorado.edu



Slides

https://github.com/ResearchComputing/applied_containerization_for_ml_short_course

Meet the User Support Team



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Overview

- Goals:
 - Understand containerization in HPC context
 - Learn Apptainer for ML workloads
 - Run TensorFlow and PyTorch containers
 - Submit jobs with SLURM
- Agenda:
 - Intro to containers
 - Apptainer deep-dive
 - TensorFlow & PyTorch demos
 - SLURM integration
 - Best practices & troubleshooting tips

Why Containers for ML in HPC?

- **Dependency Management**
 - Package complex ML software stacks with all dependencies.
 - Resolve version conflicts and avoid "dependency hell."
- **Portability**
 - Run the same environment across different HPC systems, local machines, or cloud platforms.
 - "Build once, run anywhere."
- **Scalability**
 - Easily scale workloads from a laptop to thousands of nodes.
 - Integrate seamlessly with schedulers like SLURM for large ML jobs.

Why Containers for ML in HPC? (cont.)

- **Reproducibility**

- Capture the exact software environment for consistent, repeatable results.

- **Performance**

- Near-native speed with minimal overhead; direct access to GPUs and high-speed interconnects.
- Optimized images can boost job throughput and resource utilization.

Containerization Concepts

What is a container?

- Container Image: A self-contained read-only file (or files) used to run the packaged application
- Container: A running instance of a container image
 - An isolated, encapsulated user-space instance.
 - Runs on a shared OS kernel but has its own:
 - Filesystem
 - Processes
 - Network interfaces (can be configured)

Containers vs. Virtual Machines (VMs)

Feature	Virtual Machine (VMs)	Containers
Isolation	Full (hardware/emulated)	Process/user-space
Guest OS	Full OS per VM	Shares host OS kernel
Setup	Slow (minutes)	Fast (seconds)
Resource Use	Heavy (more RAM/CPU)	Lightweight (minimal overhead)
Portability	Hypervisor-dependent	Build once, run anywhere
Use case	Legacy apps, OS-level isolation	ML, microservices, HPC, CI/CD



Key Components

- **Container Images**

- Read-only templates used to create containers.
- Contain application code, libraries, dependencies, and metadata.
- Examples: Docker Hub images, SIF files (Apptainer).

- **Runtime Environments**

- Software that runs containers (e.g., Apptainer, Docker).
- Manages container lifecycle, isolation, and resource allocation.

Key Components (cont.)

- **Mount Points for Data (Bind Mounts)**
 - Mechanism to make host directories/files accessible inside the container.
 - Essential for accessing datasets, scripts, and output directories.
- **Resource Allocation**
 - Containers share host resources (CPU, memory, GPUs).
 - HPC schedulers (like SLURM) manage resource allocation for containerized jobs.

Apptainer (formerly Singularity)

- Apptainer is the direct successor to Singularity.
- Requires a Linux system
- Runs without requiring root access
- Single-file SIF format is easy to transport and share with others
- Can convert existing Docker images to Apptainer images
- Apptainer comes pre-installed on all Alpine compute nodes, so no need to load any specific software modules!



Apptainer Definition Files

- A Definition File (or “def file” for short) is a set of blueprints explaining how to build a custom container.
- It includes specifics about the base OS, software to install, environment variables and other metadata.
 - https://apptainer.org/docs/user/latest/definition_files.html

```
Bootstrap: docker
From: ubuntu:20.04

%post
export DEBIAN_FRONTEND=noninteractive
apt-get update && apt-get install -y \
    python3 python3-pip python3-dev \
    libtcl8.6 libglu1-mesa libgomp1 \
    libblas3 liblapack3 libgfortran5 \
    wget git

# 1. Install OpenSeesPy via pip
pip3 install openseespy

# 2. Add the 'wrapper' for Tcl usage
echo '#!/usr/bin/python3' > /usr/local/bin/OpenSees
echo 'import openseespy.opensees as ops; import sys' >> /usr/local/bin/OpenSees
echo 'if len(sys.argv) > 1: ops.source(sys.argv[1])' >> /usr/local/bin/OpenSees
chmod +x /usr/local/bin/OpenSees

%environment
export LC_ALL=C
export PATH=/usr/local/bin:$PATH

%runscript
exec OpenSees "$@"
```

Working with Apptainer on CURC

- View list of Apptainer commands.

```
$ apptainer --help
```

- Look at CURC's collection of pre-built containers.

```
$ echo $CURC_CONTAINER_DIR  
$ ls $CURC_CONTAINER_DIR
```

- By default, the cache directory for Apptainer builds is **/scratch/alpine/\$USER**

```
$ echo $APPTAINER_CACHEDIR
```

*Pro Tip: change [APPTAINER_CACHEDIR](#) and [APPTAINER_TMPDIR](#) to **\$SLURM_SCRATCH** for faster builds.*

Working with Apptainer on CURC

- On CURC systems, a running container automatically bind mounts these paths: /home/\$USER, \$PWD.
- To bind any additional folders/files to your container, use the -B flag in your apptainer run, exec, and shell commands:

```
$ apptainer run -B /source/directory:/target/directory sample-image.sif  
  
$ apptainer run -B /projects/$USER,/pl/active,/scratch/alpine/$USER sample-image.sif
```

Useful Commands

<code>apptainer inspect</code>	<code>#See labels/environment vars, run scripts</code>
<code>apptainer pull</code>	<code>#pull an image from hub</code>
<code>apptainer exec</code>	<code>#Execute a command to your container</code>
<code>apptainer run</code>	<code>#Run your image as an executable</code>
<code>apptainer build</code>	<code>#Build a container</code>
<code>apptainer shell</code>	<code>#Access the command line of your container</code>

Using images from a pre-built container

- You can fetch container images from container registries such as:
 - [Docker Hub](#)
 - [NVIDIA NGC Catalog](#)
- Let's walk through an example of using pre-built Tensorflow container to train a classification model....

Start an interactive testing job on a GPU node:

```
$ sinteractive --partition=aa100 --qos=gpu-testing --nodes=1  
--ntasks=4 --time=1:00:00 --gres=gpu:a100_3g.20gb:1
```

Getting Started with TensorFlow

```
# Export paths
$ export IMAGES=/projects/$USER/containers/
$ export WORKDIR=/projects/$USER/containers/ml_work
$ mkdir -p $IMAGES $WORKDIR && cd $WORKDIR

# Pull TensorFlow container (GPU version)
$ export APPTAINER_IGNORE_PROOT=1 # temporary workaround for bug
$ apptainer pull $IMAGES/tensorflow-2.15.0-cuda12.8.sif
docker://tensorflow/tensorflow:2.15.0-gpu

# Inspect container metadata
$ apptainer inspect $IMAGES/tensorflow-2.15.0-cuda12.8.sif
```

Binding Paths & Running TF Script

- Bind mount your script directory.
- Create `mnist_tf.py` that builds and trains a simple image classification model
- `--nv`: Enables NVIDIA GPU access.

```
# Bind paths
```

```
$ apptainer exec -B $WORKDIR:/work,$IMAGES:/images $IMAGES/tensorflow-2.15.0-cuda12.8.sif ls /work
```

```
# Run the script
```

```
$ apptainer exec --nv -B $WORKDIR:/work $IMAGES/tensorflow-2.15.0-cuda12.8.sif python3 $WORKDIR/mnist_tf.py
```



Sample TensorFlow Script: `mnist_tf.py`

```
import tensorflow as tf
mnist = tf.keras.datasets.mnist
(x_train, y_train), _ = mnist.load_data()
x_train = x_train / 255.0
model = tf.keras.models.Sequential([
    tf.keras.layers.Flatten(input_shape=(28,28)),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dropout(0.2),
    tf.keras.layers.Dense(10, activation='softmax')
])
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy',
metrics=['accuracy'])
model.fit(x_train, y_train, epochs=1)
print("Done training.")
```



Access container interactively

- Shell into it

```
$ apptainer shell --nv $IMAGES/tensorflow-2.15.0-cuda12.8.sif
```

- Then run

```
$ python3
>>> import tensorflow as tf
>>> tf.__version__
>>> tf.config.list_physical_devices('GPU')
```

If it returns a list (e.g. [PhysicalDevice(name='/physical_device:GPU:0', ...)]), then TensorFlow successfully detected the GPU inside the container!

SLURM Integration: `run_mnist_tf.sh`

```
#!/bin/bash
#SBATCH --gres=gpu:a100_3g.20gb:1
#SBATCH --partition=aa100
#SBATCH --ntasks=4
#SBATCH --nodes=1
#SBATCH --qos=gpu-testing
#SBATCH --time=1:00:00
#SBATCH --job-name=tf-classify
#SBATCH --output=tf-classify.%j.out
#SBATCH --mail-type=ALL
#SBATCH --mail-user=<your email>

export IMAGES=/projects/$USER/containers/sif
export WORKDIR=/projects/$USER/containers/ml_work
cd $WORKDIR

apptainer exec --nv -B $WORKDIR:/work $IMAGES/tensorflow-2.15.0-cuda12.8.sif
python3 $WORKDIR/mnist_tf.py
```



Getting Started with PyTorch

```
# Start an interactive job on a GPU testing node
$ sinteractive --partition=aa100 --qos=gpu-testing --nodes=1 --ntasks=4 --
time=1:00:00 --gres=gpu:a100_3g.20gb:1

# Export paths
$ export IMAGES=/projects/$USER/containers/
$ export WORKDIR=/projects/$USER/containers/ml_work
$ mkdir -p $IMAGES $WORKDIR && cd $WORKDIR

# Pull TensorFlow container (GPU version)
$ export APPTAINER_IGNORE_PROOT=1 # temporary workaround for bug
$ apptainer pull $IMAGES/pytorch-2.9.0-cuda12.8.sif docker://pytorch/pytorch:2.9.0-
cuda12.8-cudnn9-runtime
```



Run PyTorch Container

Build and train a small neural network with the code [polyfit.py](#):

```
import torch
x = torch.linspace(-1, 1, 100).unsqueeze(1)
y = 3 * x ** 2 + 2 * x + 1 + 0.1 * torch.randn(x.size())
model = torch.nn.Sequential(torch.nn.Linear(1, 10), torch.nn.ReLU(),
                             torch.nn.Linear(10, 1))
loss_fn = torch.nn.MSELoss()
optimizer = torch.optim.Adam(model.parameters())
for epoch in range(100): optimizer.zero_grad(); loss_fn(model(x), y).backward();
optimizer.step()
print("Done training.")
```

Run [polyfit.py](#):

```
$ aptainer exec --bind $WORKDIR:/work $IMAGES/pytorch-2.9.0-cuda12.8.sif
python3 $WORKDIR/polyfit.py
```



Troubleshooting Common Issues

- Check GPU access:
 - `apptainer exec --nv $IMAGES/tensorflow-2.15.0-cuda12.8.sif python3 -c "import tensorflow as tf; print(tf.config.list_physical_devices('GPU'))"`
- Environment variables:
 - Pass with `--env` or set inside the container.
- Resource monitoring:
 - Use `htop`, `nvidia-smi`, or `squeue` to monitor jobs.

Best Practices: Container Management

- Version Control & Tagging
 - Tag containers with specific versions (e.g., myimage:1.0.0, myimage:latest-cuda11.8).
 - Store definition files (.def, Dockerfiles) in version control (Git).
- Data Management
 - Use bind mounts for large datasets to avoid including them in images.
 - Keep images small and focused on software environment.
- Resource Allocation
 - Match container resource needs to SLURM (or other scheduler) requests.
 - Avoid over-subscribing resources.



Any Questions